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RESEARCH ARTICLE

Intraday Trading Volume Patterns of Market Index ETFs

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ABSTRACT

Using minute-by-minute trading volume data from five ETFs (SPY, DIA, IWM, DDM, and DOG) over the period from 2013 to 2021, this paper examines the intraday trading puzzle and uncovers patterns beyond the typical U-shape. Even after controlling for potential spurious effects—such as market open/close and Wednesdays—we find persistent, albeit with reduced magnitude, statistically significant trading volume spikes at 5-minute intervals across all ETFs. Hence, while open and close, as well as Wednesdays, contribute to the pattern, they do not fully explain it. The spikes are more pronounced during negative return periods, and our models suggest increased randomness in the distribution of trading activity over time. While these predictable spikes may stem from algorithmic trading biases, we propose they could be a legacy of older algorithms influencing newer ones as they feed from historical data. The results suggest that a puzzling trading pattern persists even for highly traded ETFs; hence, it provides important findings for investors exposed to these securities.

1 | Introduction

Knowledge of trading volume dynamics in the stock market is crucial for market participants, as the presence of patterns can signal optimal times to enter or exit the market and provide insights into the strength of certain trends. Studies have documented such patterns at both intraweek and intraday frequencies. For example, Jain and Joh (1988) show that trading volumes on the New York Stock Exchange are highest on Wednesdays and lowest on Mondays and Fridays, a pattern that persists across all hours of the day. French and Roll (1986) find that volatility is higher during exchange trading hours compared to nontrading hours, illustrating the relationship between trading volume and price changes. Furthermore, the literature highlights an intraday U-shaped trading volume pattern across different markets and instruments, with higher volumes during the market open and close, and lower volumes during the midday (e.g., Abhyankar et al. 1997; Blau et al. 2009; Broussard and Nikiforov 2014; Giudici and Hu 2019; Hussain 2011; Lee et al. 2001; Y.-H. Wang and Y.-Y. Wang 2016). There is general consensus that part of these patterns is driven by the release of news. For example, corporate news, such as earnings announcements, is typically released after the close, and macroeconomic news[1] is announced before the market opens or during the first opening hour, driving the opening volume. Further, lunch breaks cause a slowdown, while portfolio adjustments and rebalancing by institutional investors trigger higher volumes during the closing minutes.

Studies that have investigated intraday trading, however, find periodicities in intraday trading volume that go beyond such U-shape patterns; thus, they offer less obvious explanations and are rather puzzling. For example, Broussard and Nikiforov (2014) study a basket of 36 US stocks and 3 ETFs over the period from 2003 to 2006 and find trading volume spikes at intervals of 5 and 10 min throughout the trading day. They suggest that this behavior could be linked to a bias in coding algorithmic trading programs, where programmers instruct the software to trade at the market open and every 5 min thereafter. Muravyev and Picard (2022) use

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transaction-level data on 120 stocks from 2008 to 2009 and show that there are spikes in trading activity at each round second and in the first 100 ms following each round second. They conclude that the increase in activity is due to a bias in coding trading algorithms: programs execute orders in loops of 1 s.

The purpose of this paper is to shed more light on the puzzling findings of intraday trading activities that go beyond the commonly observed U-shape pattern. Specifically, we examine trading volume to test whether the anomaly documented by Broussard and Nikiforov (2014) is a spurious result or if it is indeed persistent in the trading of securities. To accomplish this goal, we restrict our observations to the largest ETFs—SPDR S&P 500 ETF Trust (SPY), SPDR Dow Jones Industrial Average ETF Trust (DIA), and iShares Russell 2000 ETF (IWM)—and their leveraged counterparts—ProShares Short Dow30 (DOG) and ProShares Ultra Dow30 (DDM). The choice of these securities is deliberate: first, their trading volumes are very large, and missing data is unlikely when sampled at high frequency. Second, while ETF trading includes noise traders, it is dominated by institutional traders and authorized participants who attempt to exploit any discrepancy between the price of the ETF and its net asset value (NAV). This trading is likely to occur when there is a mispricing, rather than at a predetermined time. As such, it is expected to bias the behavior of trading volume against finding any periodicity in trading.

Our study differs from Broussard and Nikiforov (2014) in several ways. First, we use a much longer and more recent dataset, spanning 19 years and ending in 2021, while Broussard and Nikiforov (2014) cover 882 trading days from April 2003 to October 2006. Second, Broussard and Nikiforov (2014) aggregate the raw trading volumes of 36 stocks and three ETFs. Their results could potentially be driven by certain stocks or by the ETFs themselves. In contrast, we restrict our study to ETFs and examine them both individually and in aggregate. Analyzing ETFs individually allows us to investigate potential differences based on market capitalization, comparing portfolios of large stocks (SPY and DIA) with smaller ones (IWM). Additionally, we extend the study to two popular leveraged ETFs (DDM and DOG), which are composed of both stocks and derivatives. Finally, we test several models, progressively removing potential spurious factors that could explain certain intraday periodicities, such as the open and close, Wednesdays, and by controlling for positive versus negative returns. The latter might be important, as the literature suggests a relationship between stock returns and trading volume (Karpoff 1987; Chen 2012; Gebka and Wohar 2013; Bouattour and Miloudi 2023). For example, Bouattour and Miloudi (2023) show that trading volume increases during bullish periods, which is potentially brought about by overconfident investors (Daniel et al. 2002).

We find statistically significant evidence of recurring trading volume spikes at 5-min intervals across all five ETFs. Further, these spikes are, in some instances, more pronounced and significant during the opening and closing minutes, as well as on Wednesdays. Excluding the opening and closing hours, along with Wednesdays, reduces the explanatory power of our models, suggesting that some of the trading volume spikes are indeed driven by the higher trading activity during these periods. However, even under those restrictions, the 5-min spikes in trading volume do persist.

A more detailed estimation, however, indicates that these patterns are not uniform across all ETFs. For example, in the case of the S&P 500 (SPY), the trading volume spikes are statistically significantly lower at the 31st minute during open and close (9:30 a.m. and 3:30 p.m.) compared to the 31st minutes during the rest of the trading day. This suggests that, at this interval, the opening and closing periods do not contribute to volume spikes in this instance.

On the other hand, after removing the effects of the opening and closing periods, we find statistically significant positive coefficients on the Wednesday interaction term at the 1st minute. This pattern may be attributed to the Federal Open Market Committee (FOMC) announcements, which typically occur on Wednesdays at 2:00 p.m.

Most importantly, while the trading volume spikes of the non-leveraged ETFs differ only marginally between positive and negative returns, the spikes for the leveraged ETFs (DDM and DOG) are mainly explained by negative returns. This is an important finding, which, to our knowledge, is novel to literature and may, in the case of the DDM, support the overconfidence hypothesis.

Finally, while our analysis shows that the 5-min trading patterns can be partially explained by the open and close, Wednesday, and the sign of the returns, the patterns persist—albeit with a reduced magnitude—with the most pronounced trading volume spikes occurring at the half hour (Minute 31) and full hour (Minute 01) mark. Finding such patterns for ETFs is particularly puzzling, as large trades in these securities should be related to institutional traders exploiting mispricing due to discrepancies between the price of the ETF and NAV, rather than trading at specific times. This suggests either that our study has overlooked other important trading events or factors influencing volume at 5-min intervals, or that it contributes to the literature by supporting the notion that increased trading volume is driven by algorithmic trading, where human traders exhibit a behavioral bias to transact at round times, particularly at half-hour and full-hour marks. In any case, this study provides valuable insights for investors by identifying favorable trading times for entering and exiting positions due to higher liquidity and potential price movements. Moreover, these optimal trading times appear to hold across various ETF types, including large-cap ETFs such as SPY and DIA, small-cap ETFs like IWM, and leveraged ETFs, and remain consistent across different periods. However, for leveraged ETFs, the pattern appears to be more prevalent during periods of negative returns.

The remainder of the paper is organized as follows: In Section 2, we present the data as well as the employed methodology. Section 3 discusses our findings, and Section 4 concludes the paper.

2 | Data and Methodology

2.1 | Data

Our dataset comprises the trading volume recorded at each minute of the trading day of five ETFs: SPY, DIA, IWM, DOG,

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and DDM. The first three are among the largest ETFs and should have experienced nearly continuous trading throughout the day, minimizing missing data issues and eliminating volume clustering. Furthermore, SPY and IWM are composed of a very large basket of securities and follow the S&P 500 and Russell 2000, respectively, while DIA consists of only the 30 stocks included in the Dow Jones Industrial Average. Lastly, DOG and DDM are leveraged ETFs: DOG provides inverse (−1×) exposure to the DIA, whereas DDM seeks to achieve twice (2×) DIA's daily returns.

The selection of data, while restricted to ETFs, should ensure sufficient variety to make the study generalizable. Moreover, while the datasets of the individual ETFs share a common sample from January 2013 to December 2021, they each cover different overall time windows: SPY from January 2002 to December 2021, the DIA from January 2007 to December 2021, the IWM from January 2013 to December 2021, the DOG from January 2007 to December 2021, and the DDM from January 2008 to December 2021. All the datasets are obtained from TickData.

Table 1 reports descriptive statistics of the trading volumes occurring each minute in the sample. The full sample (Panel A), covering 2002 to 2021, indicates that the SPY is by far the largest by trading volume, with an average of 258,366 and a median of 140,514 shares traded in each minute. The IWM follows with an average of 66,927 and a median of 39,682. The leveraged ETFs have a considerably smaller trading volume, with averages of 5047 and 2,520 shares for the DDM and DOG, respectively. By restricting the sample from 2013 to 2021 (the common sample), the relative size of the trading volumes of the ETFs remains unaltered, although there is a decline in trading volume of all the funds. Differences in sample size are due to missing data due to inactivity in a particular minute.

The descriptive data highlights a potential problem: if we were to aggregate the volumes of all the ETFs, the behavior of the SPY would prevail over all the other funds combined. To facilitate comparisons across ETFs and the daily fluctuations in volume reported in the literature, for each day we divided the trading volume of each minute by the total trading volume of the day. The resulting value indicates what fraction of the total daily volume is traded in that minute. We call this "percentage of daily trading volume." This scale free number can be used to aggregate the trading volume across the funds, despite their considerable differences in size. Additionally, scaling the volume by the daily total would remove variability in volume associated with the day of the week, as reported by Jain and Joh (1988).

The statistics on the percentage of daily trading volume, reported to the right half of Table 1, indicate that the average trading minute results in 0.257% of the daily trading for the IWM and 0.401% of the DOG, with the other funds in between. These values are even more similar for the common sample 2013–2021, reported in Panel B. Examining the various percentiles in the common sample suggests that the leveraged ETFs experience more irregular trading than the traditional ETFs, with a 25th percentile of trading volume of 0.035% (DOG) and 0.054% (DDM) versus a range of 0.107% (SPY) to 0.072% (DIA). The leveraged ETFs are also more likely to have spikes in volume than the traditional ETFs. In fact, for the traditional ETFs, the 90th percentile of the concentration of trading volume in a given minute is about 0.5% of the daily activity. For the leveraged ETFs, it is about 0.8%. For the traditional ETFs, these 90th percentiles of trading volume are about three times their medians, while for the leveraged ETFs, they are 6–7 times.

Panel C reports the statistics for the Minutes 1, 6, 11, 16, 21, 26, 31, 36, 41, 46, 51, and 56 of each trading hour. These are the minutes that Broussard and Nikiforov (2014) identified as being responsible for the spikes. For the full sample, the average percentage of daily trading volume for those selected minutes is higher than the counterpart for all the trading minutes. This is also true for the 10th, 25th, 50th, 75th, and 90th percentiles for all the ETFs. While not a statistical test, this observation indicates that, at least at a glance, those selected minutes appear to have a distribution that is "shifted to the right" compared to the rest. Before discussing the last panel of Table 1, we need to understand how the trading volume varies throughout the day.

Figure 1 depicts the average and the 90th percentile of the percentage of daily trading volumes for each minute of the trading day. The two figures are obtained by combining the two leveraged ETFs (Figure 1B) and the three traditional ETFs (Figure 1A). The salient aspect of both graphs is the remarkable concentration of trading volume around open and close. The three traditional ETFs also display a discernible set of spikes at regular intervals that are present both in the average and in the 90th percentile of trades. The leveraged ETFs have a similar pattern, but the spikes are less pronounced. The two figures lead to two observations: first, the significant activity at open (at 9:30) and close (16:00) could be responsible for the finding of the spike in volume at Minutes 31 and 01. Second, the daily trading volume has a steady declining trend from open to about midday and then an increasing trend from midday to close. This should lead to some positive autocorrelation in the data.

In light of Figures 1A and 1B, we restricted the sample to the trading hours between 10:30 and 15:00, thus removing the majority of the intense activity due to opening and closing. By removing an entire set of minutes of the day (i.e., 1 through 60 at open and at close), we should not bias the dataset.

Panel D of Table 1 reports the descriptive statistics for this restricted sample that omits the first and last hour of trading activity. With the full sample of data, removal of the open and close leads to a subsample with a lower average trading volume than the unrestricted full sample. This is true also for the median and the percentiles. A similar finding occurs for the percentage of daily trading volume. This suggests that, at a minimum, the early and late trading of the day could have biased findings of periodicity spikes in the literature, particularly as found in the whole hour (near close) and the half hour (at open).

2.2 | Methodology

The literature and the descriptive statistics highlighted several potential issues with the data. First, aggregation cannot be done unless the data is properly scaled, due to (a) variability in fund size and (b) trading activity varying considerably by day of the week. Second, changes in volume from a given minute to the next are likely to be positively autocorrelated because volume decreases

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TABLE 1 | Distribution of trading volume and percentage of daily trading volume.

	Volume					Percentage of daily volume				
	SPY	**DIA**	**IWM**	**DDM**	**DOG**	**SPY**	**DIA**	**IWM**	**DDM**	**DOG**
Panel A: Full sample	*2002*	*2007*	*2013*	*2008*	*2007*	*2002*	*2007*	*2013*	*2008*	*2007*
Average	258,366	19,522	66,927	5,047	2,520	0.257%	0.258%	0.257%	0.322%	0.401%
10th Percentile	31,133	1,419	11,523	100	100	0.059%	0.036%	0.056%	0.025%	0.017%
25th Percentile	64,745	3,320	20,519	300	200	0.101%	0.074%	0.093%	0.061%	0.045%
Median	140,514	8,346	39,682	1,050	712	0.177%	0.153%	0.165%	0.151%	0.132%
75th Percentile	306,394	20,600	78,533	3,651	2,200	0.312%	0.301%	0.300%	0.349%	0.361%
90th Percentile	605,534	46,505	145,452	11,130	5,589	0.521%	0.547%	0.523%	0.724%	0.878%
maximum	31,151,693	5,644,780	14,068,293	734,843	1,648,109	29.965%	53.096%	28.572%	42.436%	70.328%
Standard Deviation	362,186	37,191	96,735	14,793	8,824	0.297%	0.422%	0.340%	0.631%	1.109%
# of Observations	1,957,580	1,465,458	881,202	1,093,489	941,062	1,957,580	1,465,458	881,202	1,093,489	941,062
Panel B: Common sample 2013 - 2021										
Average	204,374	11,514	66,927	1,495	2,783	0.257%	0.258%	0.257%	0.350%	0.385%
10th Percentile	40,713	1,126	11,523	100	75	0.069%	0.036%	0.056%	0.023%	0.010%
25th Percentile	68,884	2,492	20,519	200	200	0.107%	0.072%	0.093%	0.054%	0.035%
Median	127,378	5,746	39,682	501	704	0.176%	0.146%	0.165%	0.140%	0.112%
75th Percentile	242,270	12,877	78,533	1,452	2,331	0.299%	0.292%	0.300%	0.356%	0.331%
90th Percentile	438,022	25,751	145,452	3,520	6,219	0.499%	0.540%	0.523%	0.808%	0.848%
maximum	23,934,399	4,947,406	14,068,293	333,488	1,648,109	21.693%	53.096%	28.572%	42.436%	70.328%
Standard Deviation	264,816	22,272	96,735	3,588	10,389	0.311%	0.457%	0.340%	0.743%	1.164%
# of Observations	881,815	880,204	881,202	647,900	588,447	881,815	880,204	881,202	647,900	588,447
*Panel C: Minutes 1 + n*5 with n = 0 to 11 full sample*										
Average	273,823	21,311	70,530	5,471	2,874	0.273%	0.288%	0.272%	0.370%	0.456%
10th Percentile	33,178	1,561	12,250	100	100	0.063%	0.040%	0.060%	0.027%	0.017%
25th Percentile	68,694	3,629	21,809	300	204	0.107%	0.081%	0.099%	0.065%	0.048%
Median	149,100	9,100	42,203	1,100	800	0.187%	0.165%	0.176%	0.163%	0.143%
75th Percentile	325,967	22,583	84,438	3,949	2,400	0.331%	0.328%	0.321%	0.379%	0.395%
90th Percentile	650,213	51,019	156,537	12,137	6,294	0.564%	0.612%	0.569%	0.815%	0.998%
maximum	31,151,693	3,824,827	8,104,317	722,672	1,105,917	21.288%	44.320%	23.761%	37.822%	64.668%
Standard Deviation	379,185	39,634	94,089	15,968	9,878	0.300%	0.487%	0.334%	0.786%	1.253%
# of Observations	391,540	293,217	176,260	223,028	191,564	391,540	293,217	176,260	223,028	191,564
*Panel D: Minutes 1 + n*5 with n = 0 to 11 restricted to 10:30 to 15:00*										
Average	210,749	15,880	53,997	4,196	2,026	0.209%	0.205%	0.206%	0.255%	0.318%
10th Percentile	27,428	1,260	10,498	100	100	0.053%	0.033%	0.052%	0.022%	0.014%
25th Percentile	56,075	2,865	18,106	250	200	0.089%	0.065%	0.084%	0.052%	0.039%
Median	118,458	6,881	33,542	900	600	0.149%	0.128%	0.141%	0.124%	0.109%

(Continues)

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TABLE 1 | (Continued)

	Volume					Percentage of daily volume				
	SPY	DIA	IWM	DDM	DOG	SPY	DIA	IWM	DDM	DOG
75th Percentile	249,083	16,531	63,522	3,009	1,800	0.248%	0.240%	0.241%	0.276%	0.288%
90th Percentile	489,887	37,100	114,139	9,251	4,500	0.406%	0.428%	0.403%	0.565%	0.675%
Maximum	31,151,693	3,824,827	8,104,317	517,763	697,931	21.288%	26.559%	23.761%	23.943%	64.668%
Standard Deviation	298,630	30,933	75,955	12,363	6,804	0.239%	0.324%	0.267%	0.483%	0.945%
# of Observations	271,152	202,970	122,063	147,254	122,104	271,152	202,970	122,063	147,254	122,104

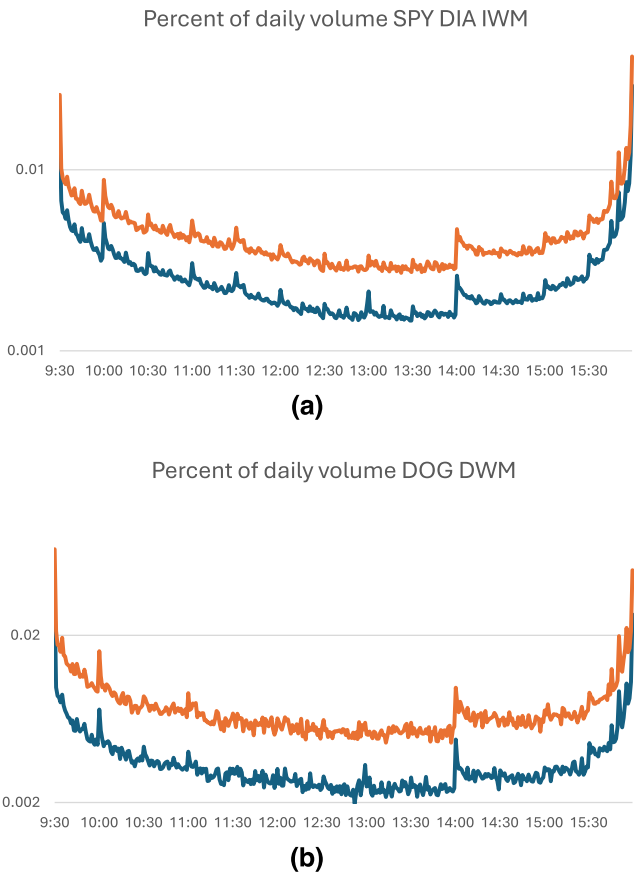

FIGURE 1 | (A) Distribution of the percentage of trading volume through the trading day. The analysis combines the three nonleveraged ETFs over the period 2013–2021, and sampling is conducted at the minute level. The lower line is the average trading volume as a percentage of the daily activity, while the upper line is the 90th percentile. The graph is in logarithmic form and shows (a) the U-shaped distribution of the trading activity and (b) the presence of spikes at regular intervals. (B) Distribution of the percentage of trading volume through the trading day. The analysis combines the two leveraged ETFs over the period 2013–2021 and sampling is conducted at the minute level. The lower line is the average trading volume as a percentage of the daily activity, while the upper line is the 90th percentile. The graph is in logarithmic form and shows (a) the U-shaped distribution of the trading activity and (b) the presence of spikes at regular intervals.

for the first half of the day and then tends to increase for the remaining half.

In order to estimate the daily volume by minute, we employ the following model:

$$VOL_t = \alpha\, VOL_{t-1} + \sum_{\min=0}^{59} \beta_{min}\, M_{min} + \varepsilon_t, \tag{1}$$

where VOL_t represents the percentage of daily trading volume at time t and $M_{\min}$ is a dichotomous variable that takes a value of 1 at minute *min* of each of the hours in the trading day (i.e., M_1 takes a value of 1 at 10:01, 11:01, etc.). Using the percentage of daily trading volume instead of the actual trading volume ensures that the volumes are equally scaled, and including the autoregressive term captures the serial correlation due to the daily trend in trading activity.

The second model is similar to the specification used by Broussard and Nikiforov (2014). In their paper, they find that not all 5-min intervals have the same spike in trading volume and separate them into nonoverlapping intervals at the round hour, the half-hour, multiples of 15 min, multiple of 10 min, and multiple of 5 min. Our model is similar to theirs but includes an autoregressive term and combines 10-min intervals with the multiples of 5 min:

$$\begin{aligned} VOL_t &= \alpha\, VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t \\ &\quad + \beta_4 MIN_16_46_t + \beta_5 MIN_k5_t + \varepsilon_t, \end{aligned} \tag{2}$$

where VOL_t is the percentage of daily trading volume at time t, MIN_01_t and MIN_31_t are dummy variables that take the value of 1 if the time interval ends in Minutes 1 and 31, respectively (i.e., hh:01 and hh:31), $MIN_16_46_t$ is a dummy variable that takes a value of 1 at Minutes 16 and 46, and MIN_k5_t is a dummy variable that takes a value of 1 at Minutes hh:06, hh:11, hh:21, hh:26, hh:36, hh:41, hh:51, and hh:56.

Finally, in Models 3a, 3b, and 3c we test if the spikes in volume are driven by specific conditions:

$$\begin{aligned} VOL_t &= \alpha\, VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t \\ &\quad + \beta_4 MIN_16_46_t + \beta_5 MIN_K5_t + O_C,_t\, (\beta_5 MIN_01_t \\ &\quad + \beta_6 MIN_31_t + \beta_7 MIN_16_46_t + \beta_8 MIN_K5_t) + \varepsilon_t, \end{aligned} \tag{3a}$$

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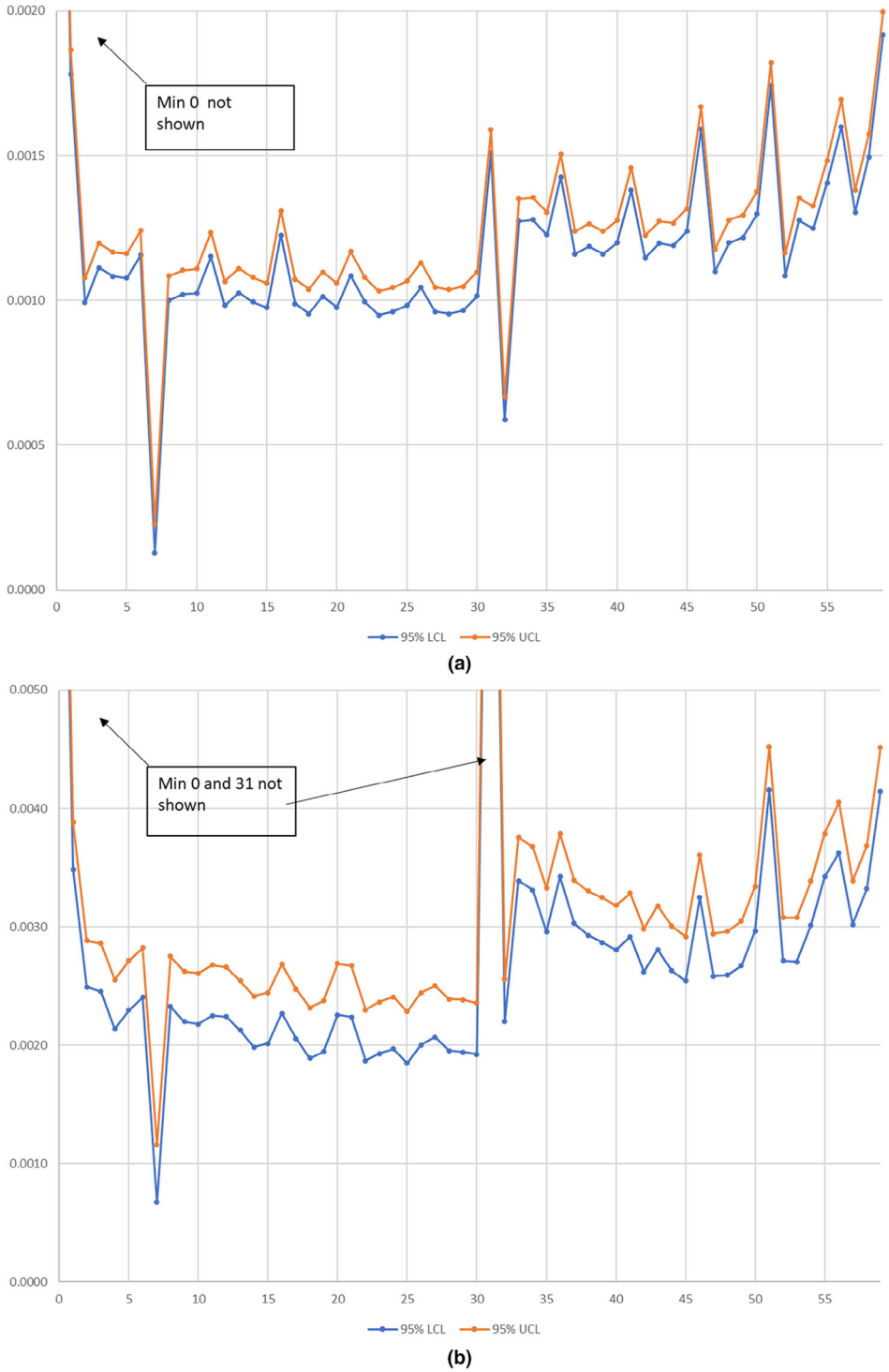

FIGURE 2 | (A) Distribution of the percentage of daily trading volume occurring at each minute of the trading day. The analysis combines the three nonleveraged ETFs for the period 2013–2021. The spikes at the whole hour, half hour, and multiples of 5 min are very pronounced. (B) Distribution of the percentage of daily trading volume occurring at each minute of the trading day. The analysis combines the two leveraged ETFs for the period 2013–2021. The spikes at the whole hour, half-hour, and multiples of 5 min are very pronounced.

$$VOL_t = \alpha\, VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t + \beta_4 MIN_16_46_t \beta_5 MIN_K5_t + WED_t\,(\beta_5 MIN_01_t + \beta_6 MIN_31_t + \beta_7 MIN_16_46_t + \beta_8 MIN_K5_t) + \varepsilon_t \quad (3b)$$

$$VOL_t = \alpha\, VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t + \beta_4 MIN_16_46_t + \beta_5 MIN_K5_t + NEG_RET_t\,(\beta_5 MIN_01_t + \beta_6 MIN_31_t + \beta_7 MIN_16_46_t + \beta_8 MIN_K5_t) + \varepsilon_t \quad (3c)$$

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where O_C_t is a dummy variable that takes a value of 1 during the opening and closing hour and zero otherwise (Model 3a), WED_t is a dummy variable that takes a value of 1 on Wednesdays and zero otherwise (Model 3b), and NEG_RET_t is a dummy variable that takes a value of 1 if the return of the ETF during time t is negative and zero otherwise (Model 3c). The rest of the variables are as defined above. The significance of the coefficients β_5, β_6, β_7, or β_8 would indicate that the periodicities in volume are more (or less) pronounced during the opening and closing minutes (Model 3a), on Wednesdays (Model 3b), or following a minute characterized by a negative return (Model 3c).

3 | Results

Our objective is to investigate whether the spikes in trading volume that occur at multiples of 5 minutes reported in the literature are not spurious. In comparison to other studies, we (a) use a much longer and more recent dataset, (b) we do not aggregate different asset classes, and (c) we control for the effect of the considerably higher trading volume near open and close. In this study we test for evidence of repeating patterns in the percentage of daily trading volume at minute intervals for major traditional ETFs (SPY, DIA, IWM) and two leveraged ETFs (DOG, DIA). To determine whether each minute of the hour is characterized by different trading activity, we estimate Model 1. The model is estimated for each ETF, for the combination of the three traditional ETFs, and for the two leveraged ETFs. Figure 2A plots the 95% confidence interval of the estimated coefficients against each corresponding minute of the hour for the SPY, DIA, and IWM combined. This graphical representation of the model allows for direct comparison of the coefficients. The 95% confidence interval of specific minutes of the hour is visibly higher (i.e., larger) than the minutes immediately before and after. If the confidence intervals of adjacent minutes are not overlapping, then we can state that they are statistically different from each other. Specifically, the 95% confidence interval of Minute 1 (off the chart) is considerably higher than Minutes 2 and 0. Similarly, Minute 31 exhibits a larger trading volume than Minutes 30 and 32. The same, albeit less pronounced, pattern can be observed at Minutes 11, 16, 21, 36, 41, 46, 51, and 56.

Figure 2B presents the 95% confidence intervals of Model 1 estimated on the combined percentage of daily trading volume of the two leveraged ETFs (DDM and DOG). While the volume spikes are less pronounced than those observed in traditional ETFs, they persist at Minutes 1 and 31 (off chart) and remain significantly larger than the immediately adjacent minutes at Minutes 36, 46, and 51.

Both estimates from Model 1 indicate that a 5-min trading pattern is most prominent at Minute 1, Minute 31, and at 5-min intervals between the half-hour and full-hour marks.

This pattern may align, at least partially, with the well-documented U-shaped or reverse J-shaped distribution of daily trading volume, which is illustrated in Figures 1A and 1B. Trading activity is typically high at the market open at 9:30 a.m. (i.e., right on the half hour), declines through midday, reaching a minimum around 1:00 p.m., and then increases again, peaking near the market close at 4:00 p.m. (right on the whole hour). As a result, identifying a repeating pattern could be heavily influenced by the disproportionately high trading activity in the opening and closing minutes. Hence, the open at 9:30 may drive the spike at the half hour and the subsequent minutes, while the spike at closing may drive the volume in the minutes leading to 4:00 p.m. (i.e., again impacting the minutes between hh:30 and the whole hour). Indeed, graphing the trading volume of our ETFs across the entire day reveals the U-shape pattern for all five ETFs.[2] To mitigate the risk of detecting spurious periodicities, we tested our models using both the entire trading day and a subset that excludes the early and late trading periods. More specifically, we estimated the model using four different samples: (a) the entire trading day with the largest available sample, (b) the largest available sample excluding the first and last hour, (c) the entire trading day with a common sample starting in 2013, and (d) the common sample with the first and last hour removed.

In Table 2, we present the R-squared and adjusted R-squared values after estimating Model 1. If the open and close of the trading day influence the volume spikes at Minutes 1 and 31, we would expect the R-squared values to be lower in the estimations that exclude the open and close periods.

Panels A and B display the explanatory power of Model 1 using the longest available sample (2002 to 2021 for the SPY) with and without the open and close periods, respectively. Table 2 shows that excluding the open and close (Panel B) significantly reduces the explanatory power of the model for all ETFs and their combination. Panels C and D, which use a common sample (2013 to 2021) across the ETFs, reach the same conclusion, suggesting that the 5-min pattern is, at least in part, driven by the opening and closing hours. Furthermore, we observe that Model (1) better explains the variability in volume for the traditional ETFs compared to their leveraged counterparts, with the highest explanatory power found for the SPY, followed by the IWM and the DIA.

Next, we test whether the repeating spikes at multiples of 5 min, detected in Figures 2A and 2B, are statistically significantly different from other minutes of the hour by estimating our regression models. Specifically, we aim to identify spikes at selected, nonoverlapping minutes: the hour (Minute 01, or hh:01), the half hour (hh:31), the quarter hours (hh:16 and hh:46), and the other 5-min multiples (hh:06, hh:11, hh:21, hh:26, hh:36, hh:41, hh:51, hh:56). To accomplish this, we estimate Model 2, which is a reduced version of Model 1. The benchmark is all other minutes of the hour, represented by the constant in the model. The results of various specifications of the model are reported in Table 3.

Table 3, Panels A and B, show that for both the longer unrestricted sample and the common sample (2013–2021), the variables are positive and statistically significant, with only four exceptions—primarily due to the DOG ETF. The findings remain consistent across different sample variants in Panels A and B, and this consistency holds whether we examine large-cap ETFs like SPY and DIA, small-cap ETFs like IWM, or leveraged ETFs such as DDM and the shorted DOG. This provides additional evidence that the selected minutes of the trading day are associated with larger trading activity than other minutes.

Given that the dependent variable is the percentage of daily trading volume, we can directly compare the coefficients. Across

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TABLE 2 | Daily trading volume by minute (Minutes 1 to 60).

	SPY	DIA	IWM	DDM	DOG	SPY/DIA/IWM	DOG/DDM
Panel A: Whole trading day, longest available sample							
Starting date	*2002*	*2007*	*2013*	*2008*	*2007*	*2013*	*2008*
R-squared	0.182	0.104	0.156	0.066	0.041	0.282	0.098
Adjusted *R*-squared	0.182	0.104	0.156	0.066	0.041	0.282	0.098
N	1957527	1463113	880985	944969	703544	879100	563805
Panel B: Open and close removed, longest available sample							
Starting date	*2002*	*2007*	*2008*	*2013*	*2013*	*2013*	*2008*
R-squared	0.077	0.031	0.033	0.026	0.014	0.081	0.029
Adjusted *R*-squared	0.077	0.031	0.033	0.025	0.014	0.081	0.029
N	1355670	1012153	610134	603449	423636	608405	323014
Panel C: Whole trading day, starting 2013							
R-squared	0.220	0.108	0.156	0.061	0.032	0.282	0.093
Adjusted *R*-squared	0.220	0.108	0.156	0.061	0.032	0.282	0.093
N	881764	879191	880985	526326	446354	879100	339774
Panel D: Open and close removed, starting 2013							
R-Squared	0.054	0.023	0.033	0.020	0.010	0.081	0.024
Adjusted *R*-squared	0.054	0.023	0.033	0.020	0.009	0.081	0.024
N	610,651	608,474	610,134	324,597	271,904	608,405	192,321

Note: This table provides the *R*-squared and the adjusted *R*-squared of Model 1: $VOL_t = \alpha\, VOL_{t-1} + \sum_{\min=0}^{59} \beta_{min}\, M_{min} + \varepsilon_t$, where VOL_t represents the trading volume measured as the percentage of the trading volume of the day that occurred at time *t*, and M_{min} is a dichotomous variable that takes a value of 1 at each minute of each of the hours in the trading day.

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TABLE 3 | Impact of Open, Close and Wednesday on Daily Trading Volume by Minute Groups.

	SPY		DIA		IWM		DDM		DOG		SPY/DIA/IWM		DOG/DDM	
	Coef.	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***
Panel A: Whole trading day, longest available sample														
	2002		2007		2013		2008		2007		2013		2008	
C	0.00147	538.5	0.00172	408.5	0.00159	35.5	0.00241	31.2	0.00326	215.9	0.0012	321.2	0.0028	244.0
VOL(-1)	0.41623	637.3	0.30339	386.3	0.37314	374.3	0.22248	226.3	0.15989	139.6	0.5070	545.6	0.2452	193.6
MIN_01	0.00057	36.4	0.00084	31.0	0.00036	13.1	0.00107	21.1	0.00050	4.9	0.0006	27.2	0.0010	13.0
MIN_31	0.00018	12.5	0.00171	68.1	0.00016	6.3	0.00419	9.4	0.00671	74.4	0.0003	14.7	0.0067	105.3
MIN_16_46	0.00020	18.8	0.00029	15.6	0.00011	5.8	0.00026	7.3	0.00000	−0.1	0.0002	15.2	0.0001	2.9
MIN_06_11_21_26_36_41_51_56	0.00011	18.7	0.00016	16.1	0.00009	9.4	0.00024	12.6	0.00018	4.6	0.0002	2.4	0.0002	9.2
R-squared	0.174		0.098		0.140		0.062		0.036		0.259		0.086	
Adjusted R-squared	0.174		0.098		0.140		0.062		0.036		0.259		0.086	
Observations	1957527		1463113		880985		944969		703544		879100		563805	
Panel B: Whole trading day, 2013 to 2021														
C	0.00139	338.0	0.00172	298.3	0.00159	35.5	0.00272	218.7	0.00324	163.3	0.0012	321.2	0.0029	181.7
VOL(-1)	0.44483	459.6	0.29946	295.4	0.37314	374.3	0.20017	151.3	0.13182	91.4	0.5070	545.6	0.2250	137.0
MIN_01	0.00057	23.5	0.00071	18.8	0.00036	13.1	0.00091	11.1	0.00045	3.4	0.0006	27.2	0.0008	7.5
MIN_31	−0.00009	−4.1	0.00220	63.0	0.00016	6.3	0.00534	72.1	0.00655	55.3	0.0003	14.7	0.0073	81.6
MIN_16_46	0.00020	12.2	0.00028	1.7	0.00011	5.8	0.00021	3.7	0.00000	0.0	0.0002	15.2	0.0001	1.4
MIN_06_11_21_26_36_41_51_56	0.00015	17.2	0.00018	13.3	0.00009	9.4	0.00032	1.6	0.00023	4.5	0.0002	2.4	0.0003	8.7
R-squared	0.197		0.098		0.140		0.053		0.026		0.259		0.077	
Adjusted R-squared	0.197		0.098		0.140		0.053		0.026		0.259		0.077	
Observations	881764		879191		880985		526326		446354		879100		339774	
Panel C: Dummy Open Close, 2013 to 2021														
C	0.00141	336.1	0.00178	31.4	0.00161	353.0	0.00279	226.5	0.00329	166.7	0.00126	325.4	0.00303	191.4
VOL(-1)	0.44011	434.0	0.27561	269.4	0.36376	353.5	0.18188	138.1	0.11985	83.2	0.49312	506.5	0.19700	120.4
MIN_01	0.00028	9.4	0.00026	5.7	0.00005	1.5	0.00016	1.6	−0.00005	−0.3	0.00028	10.5	0.00003	0.2
MIN_01*O_C	0.00087	17.2	0.00131	16.8	0.00091	15.9	0.00196	11.8	0.00127	4.7	0.00093	20.4	0.00175	8.6
MIN_31	0.00011	4.1	0.00003	0.6	0.00001	0.4	−0.00032	−3.5	−0.00010	−0.7	0.00013	5.3	−0.00029	−2.5
MIN_31*O_C	−0.00064	−12.6	0.00789	102.8	0.00064	11.1	0.01555	103.3	0.01740	72.7	0.00079	17.0	0.01737	99.2
MIN_16_46	−0.00022	−11.3	−0.00027	−8.9	−0.00027	−12.0	−0.00055	−7.8	−0.00077	−6.6	−0.00016	−9.2	−0.00075	−8.2

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TABLE 3 | (Continued)

	SPY		DIA		IWM		DDM		DOG		SPY/DIA/IWM		DOG/DDM	
	Coef.	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***
MIN_16_46*O_C	0.00138	39.2	0.00176	32.5	0.00123	3.8	0.00197	17.5	0.00199	1.8	0.00126	39.9	0.00194	14.0
MIN_06_11_21_26_36_41_51_56	−0.00026	−25.5	−0.00037	−23.0	−0.00033	−27.9	−0.00061	−16.4	−0.00075	−11.9	−0.00023	−24.6	−0.00076	−15.5
MIN_06_11_21_26_36_41_51_56*O_C	0.00134	76.2	0.00177	65.0	0.00136	68.1	0.00240	42.2	0.00247	26.0	0.00126	79.2	0.00250	35.2
R-squared	0.204		0.114		0.146		0.076		0.040		0.266		0.106	
Adjusted R-squared	0.204		0.114		0.146		0.076		0.039		0.266		0.106	
Observations	881764		879191		880985		526326		446354		879100		339774	
Panel D: Open, close removed, 2013 to 2021														
C	0.00144	367.8	0.00155	288.3	0.00156	351.6	0.00217	186.4	0.00250	123.0	0.00133	374.9	0.00225	148.8
VOL(−1)	0.24029	192.4	0.15713	124.6	0.19191	152.5	0.14081	83.9	0.09617	52.2	0.29141	237.0	0.15431	7.1
MIN_01	0.00063	27.7	0.00072	20.3	0.00043	16.0	0.00089	11.5	0.00080	5.8	0.00060	30.5	0.00092	9.6
MIN_31	0.00047	22.9	0.00049	15.5	0.00041	16.6	0.00040	5.8	0.00076	6.2	0.00045	25.8	0.00061	7.2
MIN_16_46	0.00012	7.8	0.00018	7.5	0.00012	6.4	0.00017	3.2	0.00008	0.9	0.00014	1.9	0.00013	2.0
MIN_06_11_21_26_36_41_51_56	0.00008	1.3	0.00009	6.8	0.00006	6.0	0.00011	3.8	0.00011	2.1	0.00008	11.6	0.00013	3.6
R-squared	0.059		0.026		0.038		0.022		0.010		0.087		0.026	
Adjusted *R*-squared	0.059		0.026		0.038		0.022		0.010		0.087		0.026	
Observations	608,384		606,207		607,867		323,116		270,569		606,138		191,289	
Panel E: Open close removed and dummy Wednesday, 2013 to 2021														
C	0.00144	367.8	0.00155	288.3	0.00156	351.6	0.00218	186.4	0.00250	123.1	0.00133	374.9	0.00225	148.8
VOL(−1)	0.24032	192.4	0.15715	124.7	0.19190	152.4	0.14080	83.8	0.09616	52.2	0.29142	237.0	0.15427	70.1
MIN_01	0.00051	20.2	0.00057	14.3	0.00040	13.0	0.00060	7.0	0.00044	2.9	0.00049	22.5	0.00060	5.6
MIN_01*WED	0.00056	10.0	0.00076	8.8	0.00019	2.9	0.00143	7.6	0.00175	5.2	0.00051	10.7	0.00153	6.6
MIN_31	0.00045	20.0	0.00047	13.1	0.00039	14.4	0.00036	4.7	0.00050	3.7	0.00043	22.1	0.00045	4.8
MIN_31*WED	0.00006	1.2	0.00014	1.8	0.00006	1.1	0.00021	1.2	0.00125	4.2	0.00009	2.1	0.00079	3.8
MIN_16_46	0.00010	5.9	0.00016	5.9	0.00012	5.9	0.00015	2.6	0.00004	0.4	0.00013	9.1	0.00010	1.4
MIN_16_46*WED	0.00009	2.5	0.00010	1.7	−0.00002	−0.4	0.00007	0.5	0.00019	0.8	0.00005	1.5	0.00014	0.9
MIN_06_11_21_26_36_41_51_56	0.00008	8.9	0.00008	5.8	0.00005	4.3	0.00010	3.2	0.00011	2.0	0.00007	9.7	0.00009	2.3
MIN_06_11_21_26_36_41_51_56*WED	0.00002	1.0	0.00003	0.9	0.00006	2.7	0.00004	0.7	0.00000	0.0	0.00003	1.8	0.00018	2.1

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TABLE 3 | (Continued)

	SPY		DIA		IWM		DDM		DOG		SPY/DIA/IWM		DOG/DDM	
	Coef.	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***
R-squared	0.059		0.026		0.038		0.022		0.010		0.087		0.026	
Adjusted *R*-squared	0.059		0.026		0.038		0.022		0.010		0.087		0.026	
Observations	608,384		606,207		607,867		323,116		270,569		606,138		191,289	
Panel F: Open, close and Wednesday removed, 2013 to 2021														
C	0.00145	364.9	0.00154	287.3	0.00156	346.2	0.00217	183.0	0.00249	121.8	0.00133	37.4	0.00225	146.6
VOL(−1)	0.23524	185.3	0.16060	125.5	0.19058	149.0	0.14065	82.4	0.09782	52.5	0.29216	234.0	0.15519	69.9
MIN_01	0.00061	26.3	0.00072	20.4	0.00043	15.4	0.00092	11.7	0.00080	5.8	0.00058	29.8	0.00093	9.6
MIN_31	0.00047	22.6	0.00050	15.7	0.00041	16.4	0.00040	5.7	0.00077	6.3	0.00045	25.7	0.00061	7.1
MIN_16_46	0.00012	7.7	0.00018	7.7	0.00012	6.3	0.00018	3.4	0.00010	1.1	0.00015	11.0	0.00015	2.2
MIN_06_11_21_26_36_41_51_56	0.00008	1.0	0.00009	6.9	0.00006	5.7	0.00011	3.9	0.00013	2.4	0.00008	11.4	0.00014	3.8
R-squared	0.057		0.027		0.037		0.022		0.011		0.088		0.026	
Adjusted *R*-squared	0.057		0.027		0.037		0.022		0.011		0.088		0.026	
Observations	589,115		587,153		588,737		312,592		261,523		587,096		184752	

Note: This table provides the estimates for selected nonoverlapping minutes for the longest available sample (Panel A), the sample from 2013 to 2021 (Panel B), the opening and closing hours removed (Panel D), and the opening/closing and Wednesdays removed. The estimations are based on Model 2: $VOL_t = \alpha\, VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t + \beta_4 MIN_16_46_t + \beta_5 MIN_K5_t + \varepsilon_t$, where VOL_t represents the trading volume measured as the percentage of the trading volume of the day that occurred at time t. MIN_01_t *and* MIN_31_t are dummy variables that take the value of 1 if the time interval ends in Minutes 1 and 31, respectively (i.e., hh:01 and hh:31). $MIN_16_46_t$ is a dummy variable that takes a value of 1 at Minutes 16 and 46, and MIN_k5_t is a dummy variable that takes a value of 1 at Minutes hh:06, hh:11, hh:21, hh:26, hh:36, hh:41, hh:51, and hh:56. Panels C and D represent a modified version of Model 2, with the intervals being interacted with a dummy variable (O_C) that takes the value of 1 if the interval occurs in the first or last trading hour (Model 3a, Panel C), or with a dummy variable (WED) that takes the value of 1 if the interval occurs on a Wednesday (Model 3b, Panel D), and 0 otherwise.

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all ETFs, the largest coefficients are either associated with minute 01 (SPY, IWM) or Minute 31 (IWM, DDM, and DOG). Perhaps this is an indication that their trading activity is strongest at open (9:30/Minute 31) or close (4:00/Minute 01).

Panel C presents the results of Model 3a, a modified version of Model 2, which estimates the restricted sample from 2013 to 2021. This model separates the 5-min intervals between the open and close (O_C) and the rest of the day using interaction terms, including a dummy variable that takes a value of 1 if the minute is in the first or last trading hour of the day (O_C). The value of each estimated coefficient of our main independent variables represents the difference in the percentage of daily trading volume occurring at that particular group of minutes, relative to all minutes of the day. Furthermore, the coefficient of the interaction term reflects the difference between a specific group of minutes and the same minutes occurring within one hour of the open and close. For example, for SPY, the positive and statistically significant coefficient at minute 01 (MIN_01) indicates statistically significant trading volume spikes around the hour, while the positive and statistically significant coefficient of the interaction term (MIN_01*O_C) suggests that trading volume at whole hours near the open and close exhibits even larger spikes.

Additionally, the results presented in Panel C show that most interaction terms are statistically significant and positive. This suggests that the spikes in trading volume are somewhat related to the first hour after opens, as well as the hour preceding the market close. Interestingly, the coefficient for the 31st minute of the SPY is negative and statistically significant, which suggests that in the case of the S&P 500 ETF, the trading volume during the opening minute at 9:30 a.m. and during the closing hour at 3:30 p.m. is lower compared to the other 31st minutes during the rest of the day. This provides some evidence against the possibility that the open and close are driving the volume spikes, at least in the case of the SPY. Most notably, across all ETFs and their aggregates, the introduction of the interaction term lowered the estimates of the trading volume spikes, with some coefficients being either not significant or even negative. For example, the coefficients on the quarter hour minutes (MIN_16_46) and the coefficients on the remaining set of 5 min (MIN_06_11_21_26_36_41_51_56) are all statistically significantly negative, yet relatively smaller than the others in magnitude. This suggests a clear difference in volume spikes during the opening and closing for the quarter hour and the remaining set of 5 min for all ETFs when compared to the same intervals for the rest of the day.

Thus, upon reviewing Panel C, one might wonder whether the observed periodic spikes in volume are primarily driven by the first and last hours of the trading day. To explore this further, we re-estimate Model 2 in Panel D, excluding the first and last trading hours and restricting the sample to trading activity occurring between 10:30 a.m. and 3:00 p.m. While excluding the open and close lowers the explanatory power of our model as well as the size of some of the coefficients, we find, with a few exceptions, statistically significant positive spikes in trading volume at multiples of 5 min across all ETFs. Minute 01 exhibits the strongest spike, followed by Minute 31, while the quarter-hour and remaining 5-min intervals show weaker but still statistically significant spikes in most cases. The only exceptions are the coefficients for MIN_06_11_21_26_36_41_51_56 for the SPY and MIN_16_46 for the DOG, which are not significant. Therefore, restricting the sample reveals that the 5-min pattern persists, albeit with a reduced magnitude, indicating that it cannot be solely attributed to the open and close.

3.1 | Wednesday Effect?

The literature suggests that there might be a specific day of the week effect when it comes to trading volume. For example, Jain and Joh (1988) investigate the trading volume at the New York Stock Exchange and find it to be highest on Wednesdays and lowest on Mondays and Fridays. Moreover, the FOMC typically concludes its meetings on Wednesdays at 2:00 p.m. ET with the release of its policy decision. Additionally, the FOMC minutes are usually released on Wednesdays at the same time. This raises the question of whether trading volume spikes are more pronounced on Wednesdays compared to other days, particularly at Minute 01.

To examine whether the pattern differs on Wednesdays, Model 3b is estimated by interacting the variables from Model 2 with a dummy variable (WED) that equals one on Wednesdays and zero otherwise. Given that the dependent variable is the percentage of daily trading volume, a positive coefficient on the interaction term would suggest an additional share of volume at that minute on Wednesdays compared to the other days.

The results are presented in Panel E. The coefficients on the Wednesday interaction terms are mostly statistically insignificant, except for Minute 01 (MIN_01WED) of each trading hour, where they are statistically significant and positive. Moreover, the coefficients for the Minute 01 interaction term (MIN_01WED) are the largest across all ETFs and their aggregates, except for the ETF representing the Russell 2000 small stock index (IWM). This implies that the general 5-min patterns detected in this paper, except for Minute 01, are not driven by the Wednesday effect. The strong and statistically significant trading spikes at Minute 01, however, may indicate that the FOMC announcement at 2:00 p.m. contributes to the observed spike at full-hour marks, though it does not fully explain it. To further investigate this issue, we estimate Model 2 on a sample that is most biased *against* detecting spikes, as it excludes both the open and close as well as Wednesdays. While the explanatory power of the model is slightly reduced compared to the version that includes Wednesdays, the coefficients for the selected 5-min intervals remain significant and positive. This suggests that although Wednesday trading activity, particularly at Minute 01 (MIN_01), contributes to the 5-min trading pattern, it is not the primary driver.

Our findings indicate that the 5-min trading pattern of the ETFs persists even after removing the effects of the open/close and Wednesdays, though at a lower magnitude. Notably, the spikes remain most pronounced and statistically significant at the half-hour (Minute 31) and full-hour (Minute 01) marks. This may suggest that these specific times offer more favorable trading opportunities for investors to enter or exit positions in these ETFs due to increased liquidity and potential price movements.

3.2 | Specific Time-Period Effect

To ensure that the observed anomaly is not driven by a specific period, we divide the sample into three distinct 3-year subperiods:

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2013–2015, 2016–2018, and 2019–2021. We then re-estimate Model 2 on the most restricted sample, which excludes the first and last trading hours as well as Wednesdays.

The findings, presented in Table 4, confirm that the 5-min trading pattern persists across all subperiods for our ETFs. However, the two leveraged ETFs exhibit the most pronounced pattern during the most recent period (2019–2021). Additionally, the stronger trading activity at the half-hour and full-hour marks remains consistent over time.

Notably, for the nonleveraged ETFs, the explanatory power is strongest in the 2013–2015 period (Panel A) but declines substantially in the second subperiod (Panel B) and third subperiod (Panel C). In contrast, for the leveraged ETFs, it increases slightly. This may be attributed to the lower overall trading volume of DOG and DDM compared to their nonleveraged counterparts. Furthermore, the nonleveraged ETFs exhibit a decline in the magnitude of the autocorrelation term. This could indicate that the trading volume has become less clustered over time. This is particularly visible in the aggregate of the nonleveraged ETFs.

Before providing a possible way to explain this decline in the 5-min pattern, we need to briefly discuss the evolution of trading algorithms. Barry Johnston (2010) describes the early algorithms as "order slicers" that simply split a large order into smaller ones spread through the day, possibly according to a predetermined schedule. While those algorithms were acceptable for concealing orders or for reducing price impact, they lent themselves to possible predatory trading and could not accomplish more sophisticated goals requiring more complex criteria for execution. The second generation of algorithms focused on the optimization of the tradeoff between price impact and execution risk: a joint goal that required a dynamic approach. Finally, the most recent algorithms are optimized to work across trading venues to seek liquidity and speed of execution, making them the most opportunistic.

Explaining the 5-min pattern as a simple bias in the coding of algorithms is highly plausible with the earliest generation, where spreading a trade through the day was the main objective. But with the advent of dynamic and opportunistic algorithms, the presence of a bias in coding should no longer be visible in the data. Perhaps the decline of the explanatory power in our models reported in Table 4 is a reflection of the increase in popularity of those more modern algorithms. However, why are the 5-min spikes then still detectable? A plausible but speculative explanation could be that the bias was introduced in the coding of early algorithms. This may cause the visible spike in volume at multiples of 5 min. Subsequent algorithms that used historical trends in volume as inputs then used the spikes to predict the larger amount of liquidity available at those times, prolonging their existence. The proliferation of the more sophisticated algorithms capable of accomplishing more sophisticated goals then slightly mitigated the effect, increasing the randomness in the timing of the trades, which is reflected in the reduction of the *R*-squared of our models.

3.3 | Negative Return Effect?

The literature suggests a causal relationship of stock returns to trading volume (Karpoff 1987; Chen 2012; Gebka and Wohar 2013). Moreover, some studies link the sign of stock returns to trading volume (e.g., Daniel et al. 2002; Kudryavtsev 2017; Bouattour and Miloudi 2023). For example, Bouattour and Miloudi (2023) find a positive relationship between stock returns and market turnover during bullish periods and a negative relationship during bearish periods. The positive relationship is attributed to investor overconfidence, which leads to increased trading, while the negative relationship suggests that investors tend to hold onto losing positions in declining markets. Consequently, this raises the question of whether the spikes observed at 5-min intervals are also influenced by the sign of returns.

To examine whether the spikes in volatility are indeed related to the sign of the returns, we estimate Model 3c, controlling for negative returns by interacting each of our main independent variables with a dummy variable (NEG_RET). The dummy variable takes a value of 1 when the return during that period is negative and zero otherwise. Since lower trading volume is generally expected during down markets, we anticipate that the coefficients on the interaction terms will be negative and statistically significant, while the remaining variables should remain positive.

Table 5 shows the results for the most restrictive case, excluding open and closed as well as Wednesdays and the common sample from 2013 to 2021. For the SPY and the DIA, the coefficients are positive and statistically significant, except for the interaction terms at Minute 01 (MIN_01) and Minute 31 (MIN_31) for the SPY and the interaction terms at Minute 31 (MIN_31) for the DIA. The coefficients on these three interaction terms are statistically insignificant, suggesting that the spikes during these intervals are not different from the positive returns. However, the positive and statistically significant coefficients for the other intervals indicate that volume spikes occur for positive returns and become even more pronounced when returns are negative. This implies that for the two large ETFs (SPY and DIA), trading volume spikes are driven by both positive and negative returns, with negative returns contributing slightly more to these spikes.

Interestingly, for the IWM, which represents the small Russell 2000 stocks, we do find that the coefficients on the interaction terms are mainly negative, yet they are statistically insignificant, indicating no difference between the spikes at positive versus negative return. Most remarkably, however, for the leveraged ETFs, the coefficients on the interaction terms are all statistically significantly positive, while the other coefficients at the 5-min periodicities are mainly insignificant or negative. This result may be related to negative returns leading to losses, potentially triggering speculative investors to experience more anxiety and prompting them to exit their positions rather than holding onto their securities, hence causing higher trading volume.

Overall, the results in Table 5 suggest that the periodic spikes occurring at 5-min intervals are relatively consistent between positive and negative returns for the nonleveraged ETFs. This finding, at least in the case of these ETFs, does not support the literature's suggestion that investors increase trading volume during periods of positive returns compared to negative ones.

On the other hand, the spikes for the leveraged ETFs (DDM and DOG) are predominantly explained by negative returns.[3] This is a crucial finding that adds new insights to the literature. This also

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TABLE 4 | Sub-periods: Impact on daily trading volume by minute groups, open, close, and Wednesdays removed.

	SPY		*DIA*		*IWM*		*DDM*		*DOG*		*SPY/DIA/IWM*		*DOG/DDM*	
	Coef.	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***	***Coef.***	***t-stat.***
Panel A: Years 2013, 2014, 2015, Open Close Wednesday Removed														
C	0.00103	185.5	0.00134	172.8	0.00133	19.4	0.00273	10.1	0.00305	57.5	0.00106	19.0	0.00284	61.0
VOL(−1)	0.44917	219.2	0.26592	122.2	0.29701	137.6	0.12282	35.4	0.09379	24.7	0.42958	209.4	0.17027	31.1
MIN_01	0.00059	2.5	0.00061	12.6	0.00045	11.0	0.00116	6.5	0.00089	2.5	0.00056	19.2	0.00116	4.0
MIN_31	0.00057	22.1	0.00060	13.7	0.00047	12.6	0.00041	2.6	0.00073	2.5	0.00054	2.5	0.00058	2.4
MIN_16_46	0.00007	3.8	0.00009	2.9	0.00011	3.8	0.00007	0.6	−0.00011	−0.5	0.00010	5.0	−0.00016	−0.8
MIN_06_11_21_26_36_41_51_56	0.00008	7.4	0.00010	5.7	0.00004	3.0	0.00017	2.7	0.00005	0.3	0.00008	7.3	0.00014	1.3
R-squared	0.200		0.073		0.089		0.017		0.009		0.186		0.027	
Adjusted *R*-squared	0.200		0.073		0.089		0.017		0.009		0.186		0.026	
Observations	195,905		195035		195931		75669		67379		195033		36297	
Panel B: Years 2016, 2017, 2018, open close Wednesday removed														
C	0.00156	214.6	0.00161	16.3	0.00171	199.4	0.00228	102.6	0.00269	68.5	0.00147	224.8	0.00249	84.1
VOL(−1)	0.18553	83.6	0.14240	63.7	0.13950	62.4	0.14966	48.6	0.09480	28.8	0.23432	106.6	0.12716	32.0
MIN_01	0.00055	12.8	0.00072	1.7	0.00033	6.2	0.00071	5.0	0.00055	2.1	0.00053	14.7	0.00066	3.6
MIN_31	0.00042	1.9	0.00044	7.3	0.00032	6.6	0.00038	2.9	0.00045	1.9	0.00039	12.0	0.00038	2.3
MIN_16_46	0.00014	4.9	0.00025	5.6	0.00008	2.3	0.00032	3.2	0.00018	1.0	0.00016	6.7	0.00027	2.1
MIN_06_11_21_26_36_41_51_56	0.00009	6.1	0.00005	2.3	0.00004	2.0	0.00009	1.6	0.00008	0.8	0.00007	5.0	0.00004	0.6
R-squared	0.036		0.021		0.020		0.025		0.011		0.057		0.020	
Adjusted *R*-squared	0.036		0.021		0.020		0.025		0.011		0.057		0.020	
Observations	196,393		195,724		196,235		94,966		77,082		195,698		51,821	
Panel C: Years 2019, 2020, 2021, open close Wednesday removed														
C	0.00155	208.5	0.00157	161.1	0.00159	204.9	0.00184	115.6	0.00205	92.5	0.00140	218.3	0.00192	11.7
VOL(−1)	0.17888	8.7	0.11929	54.0	0.16958	76.3	0.13487	52.8	0.10120	36.6	0.24789	113.5	0.15415	51.8

(Continues)

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TABLE 4 | (Continued)

	SPY		*DIA*		*IWM*		*DDM*		*DOG*		*SPY/DIA/IWM*		*DOG/DDM*	
	Coef.	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*
MIN_01	0.00066	14.8	0.00083	12.7	0.00049	1.4	0.00092	8.7	0.00089	5.9	0.00067	18.7	0.00097	8.7
MIN_31	0.00038	9.6	0.00044	7.5	0.00043	1.1	0.00040	4.1	0.00093	6.9	0.00042	13.2	0.00069	7.0
MIN_16_46	0.00014	4.8	0.00021	4.7	0.00016	5.1	0.00014	1.9	0.00014	1.3	0.00018	7.4	0.00018	2.3
MIN_06_11_21_26_36_41_51_56	0.00008	4.8	0.00011	4.6	0.00009	5.0	0.00010	2.4	0.00021	3.8	0.00010	7.5	0.00019	4.5
R-squared	0.034		0.016		0.030		0.020		0.012		0.064		0.028	
Adjusted *R*-squared	0.034		0.016		0.030		0.020		0.012		0.064		0.028	
Observations	196,817		196,394		196,571		141,957		117,062		196,365		96634	

Note: This table provides the estimates for selected nonoverlapping minutes for the three subperiods: 2013 to 2015, 2016 to 2018, and 2019 to 2021. The estimations are based on Model (2), where $VOL_t = \alpha\ VOL_{t-1} + \beta_1 + \beta_2 MIN_01_t + \beta_3 MIN_31_t + \beta_4 MIN_16_46_t + \beta_5 MIN_K5_t + \varepsilon_t$, where VOL_t represents the trading volume measured as the percentage of the trading volume of the day that occurred at time t. MIN_01_t and MIN_31_t are dummy variables that take the value of 1 if the time interval ends in Minutes 1 and 31, respectively (i.e., hh:01 and hh:31). $MIN_16_46_t$ is a dummy variable that takes a value of 1 at Minutes 16 and 46, and MIN_k5_t is a dummy variable that takes a value of 1 at Minutes hh:06, hh:11, hh:21, hh:26, hh:36, hh:41, hh:51, and hh:56.

aligns with the notion that investors in DOG, being short, may engage in overconfident trading when they stand to benefit from negative returns.

4 | Conclusions

Knowledge of trading volume is important for investors who intend to easily find a counterpart to a transaction and wish to minimize price impact. Many studies in the literature have shown that the trading volume of assets varies with the day of the week and within the day—with a U-shaped pattern. Furthermore, Broussard and Nikiforov (2014) show the presence of a puzzling intraday spike in volume that repeats at intervals of 5 min. The authors speculate that their findings are due to a bias in coding trading algorithms. Muravyev and Picard (2022) find a similar pattern repeating near the whole second and adopt the same explanation. Both authors use data from 2009 or older and conduct the analysis over a portfolio of securities with heterogeneous characteristics. Their findings could be spurious and possibly due to the specific time period or could be driven by a few securities that have an impact on the whole set of their portfolios. In this paper we re-examine this issue by using five ETFs. This choice is intentional: their trading is motivated by the needs of investors, like other stocks, but also by the activity of authorized participants, who aim to profit from discrepancies between the price and the NAV of each ETF. To the extent that mispricings occur randomly, the authorized participant's trading of the ETFs should not follow a pattern. Furthermore, we structure our study in such a way as to bias our sample against finding the pattern.

Using a sample spanning over a decade of minute-level data, we first confirm that the intraday pattern of volume is U-shaped. We use this finding to hypothesize that the possible spikes in volume that occur every 5 min could be influenced by the intense activity that occurs near open and close. Testing indicates that, while the 5-min trading spikes occur in our sample, they are more pronounced near open and close. Testing on a sample that omits open and close indicates that the 5-min pattern persists.

Following the literature that indicates heterogeneous trading across weekdays, we hypothesize that Wednesdays, the day with larger trading and a concentration of macroeconomic announcements, could be at least partly responsible for the 5-min pattern. Testing indicates that on Wednesdays the 5-min spikes persist and, for most ETFs, are more pronounced than those on other days of the week. Restricting the sample to days other than Wednesday and removing the first and last hour of trading shows a persistence of the 5-min pattern.

A dynamic examination of the pattern, obtained by splitting our sample into three 3-year subsamples, shows that while the 5-min effect persists, the explanatory power of the models declines over time for the nonleveraged ETFs. A possible explanation for this decline could be that, while a bias could have been introduced in the early schedule driven algorithms, their evolution to include the ability to dynamically analyze the market introduced more randomness in the timing of the trades.

Lastly, following the literature on the linkages between the returns and trading volumes, we test whether the 5-min pattern

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TABLE 5 | Impact of negative returns on daily trading volume by minute groups, open, close, and Wednesday removed.

2013 to 2021	**SPY**		**DIA**		**IWM**		**DDM**		**DOG**	
	Coef.	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*	*Coef.*	*t-stat.*
C	0.00147	334.5	0.00156	258.6	0.00157	314.5	0.00216	169.4	0.00249	111.2
VOL(−1)	0.22163	158.2	0.14958	105.7	0.18218	129.1	0.13749	73.1	0.09553	46.3
MIN_01	0.00050	14.4	0.00048	8.8	0.00041	9.8	0.00018	1.7	0.00019	1.1
MIN_01*NEG_RET	0.00004	0.9	0.00020	2.6	−0.00002	−0.3	0.00123	7.1	0.00102	3.0
MIN_31	0.00045	14.5	0.00046	9.5	0.00041	1.9	0.00010	1.0	0.00036	2.3
MIN_31*NEG_RET	0.00002	0.4	0.00002	0.3	−0.00001	−0.3	0.00081	5.2	0.00058	2.0
MIN_16_46	0.00004	1.9	0.00008	2.2	0.00013	4.7	−0.00020	−2.9	−0.00020	−1.7
MIN_16_46*NEG_RET	0.00013	3.7	0.00018	3.4	−0.00001	−0.3	0.00113	9.4	0.00102	4.3
MIN_06_11_21_26_36_41_51_56	0.00006	5.2	0.00004	2.1	0.00003	2.1	−0.00021	−5.7	−0.00010	−1.6
MIN_06_11_21_26_36_41_51_56*NEG_RET	0.00004	2.5	0.00010	3.7	0.00005	2.2	0.00100	16.4	0.00089	7.3
R-squared	0.051		0.023		0.034		0.022		0.010	
Adjusted *R*-squared	0.051		0.023		0.034		0.022		0.010	
Observations	485,632		483,859		485,177		257,689		215,806	

Note: This table provides the estimates for selected nonoverlapping minutes for the longest available common sample from 2013 to 2021 with Wednesdays and open / close removed. The estimations are based on Model 3c: $VOL_t = \alpha\ VOL_{t-1} + \beta_1 + \beta_2 MIN_1_31_t + \beta_3 MIN_16_46_t + \beta_4 MIN_K5_t + NEG_RET_t(\beta_5 MIN_1_t + \beta_6 MIN_31_t + \beta_7 MIN_16_46_t + \beta_8 MIN_K5_t) + \varepsilon_t$, where VOL_t represents the trading volume measured as the percentage of the trading volume of the day that occurred at time *t*. MIN_01_t *and* MIN_31_t are dummy variables that take the value of 1 if the time interval ends in Min 1 and 31, respectively (i.e., hh:01 and hh:31). $MIN_16_46_t$ is a dummy variable that takes a value of 1 at Minutes 16 and 46, and MIN_k5_t is a dummy variable that takes a value of 1 at Minutes hh:06, hh:11, hh:21, hh:26, hh:36, hh:41, hh:51, and hh:56. Finally, the intervals are being interacted with a dummy variable (NEG_RET) that takes the value of 1 if the return of the ETF during time *t* is negative and 0 otherwise.

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is related to the sign of the returns. Using the most restrictive sample, which excluded Wednesdays and the trading near open and close, we find that for the leveraged ETFs, the 5-min pattern is more pronounced when returns are negative.

Overall, the spike in trading volume that occurs at multiples of 5 min persists. Future research should examine, on a larger scale, whether this pattern has an impact on returns, volatility, and bid-ask spreads. Additionally, the issue could be investigated by examining individual trading venues rather than their aggregate volume.

Data Availability Statement

Data sharing is not applicable to this article, as no data were created or analyzed in this study.

Endnotes

[1] Nonfarm Payrolls, Consumer Price Index, Producer Price Index and Unemployment Rate are announced at 8:30 a.m.; Consumer Confidence Index, S&P/Case-Shiller Home Price Index, and FHFA House Price Index at 9:00 a.m.; and ISM Manufacturing PMI, ISM Non-Manufacturing PMI, Job Openings and Labor Turnover Survey (JOLTS), and Existing Home Sales at 10:00 a.m.

[2] The graphs of individual ETFs are not presented in this study to preserve space but are available from the authors upon request.

[3] Analyzing the different sub-periods, as done in Table 4 (not reported to preserve space), did not change these main results presented in Table 5.

References

Abhyankar, A., D. Ghosh, E. Levin, and R. J. Limmack. 1997. "Bid-Ask Spreads Trading Volume and Volatility: Intra-Day Evidence From the London Stock Exchange." *Journal of Business Finance and Accounting* 24: 306–686.

Blau, B. M., B. F. V. Ness, and R. A. V. Ness. 2009. "Intraday Stealth Trading: Which Trades Move Prices During Periods of High Volume?" *Journal of Financial Research* 32: 1–21.

Bouattour, M., and A. Miloudi. 2023. "Another Look at the Asymmetric Relationship Between Stock Returns and Trading Volume: Evidence From the Markov-Switching Model." *Review of Accounting and Finance* 23: 256–279.

Broussard, J. P., and A. Nikiforov. 2014. "Intraday Periodicity in Algorithmic Trading." *Journal of International Financial Markets, Institutions and Money* 30: 196–204.

Chen, S. S. 2012. "Revisiting the Empirical Linkages Between Stock Returns and Trading Volume." *Journal of Banking & Finance* 36: 1781–1788.

Daniel, K., D. Hirshleifer, and S. H. Teoh. 2002. "Investor Psychology in Capital Markets: Evidence and Policy Implications." *Journal of Monetary Economics* 49: 139–209.

French, K. R., and R. Roll. 1986. "Stock Return Variances: The Arrival of Information and the Reaction of Traders." *Journal of Financial Economics* 17: 5–26.

Gebka, B., and M. E. Wohar. 2013. "Causality Between Trading Volume and Returns: Evidence From Quantile Regressions." *International Review of Economics & Finance* 27: 144–159.

Giudici, E., and H. Hu. 2019. "Intraday Patterns in Trading Volume of the SPY ETF." *International Journal of Business and Social Science* 10: 91–100.

Hussain, S. M. 2011. "The Intraday Behavior of Bid-Ask Spreads, Trading Volume and Return Volatility: Evidence From Dax 30." *International Journal of Economics and Finance* 3: 23–34.

Jain, P. C., and G. H. Joh. 1988. "The Dependence Between Hourly Prices and Trading Volume." *Journal of Financial and Quantitative Analysis* 23: 269–283.

Johnson, B. 2010. *Algorithmic Trading and DMA an Introduction to Direct Access Trading Strategies*. Myeloma Press.

Karpoff, J. M. 1987. "The Relation Between Price Changes and Trading Volume: A Survey." *Journal of Financial and Quantitative Analysis* 22: 109–126.

Kudryavtsev, A. 2017. "The Effect of Stock Return Sequences on Trading Volumes." *International Journal of Financial Studies* 5: 1–15.

Lee, Y. T., R. C. Fok, and Y. J. Liu. 2001. "Explaining Intraday Pattern of Trading Volume From the Order Flow Data." *Journal Of Business Finance and Accounting* 28: 199–230.

Muravyev, D., and J. Picard. 2022. "Does Trade Clustering Reduce Trading Costs? Evidence From Periodicity in Algorithmic Trading." *Financial Management* 51: 1201–1229.

Wang, Y.-H., and Y.-Y. Wang. 2016. "The Information Content of Intraday Implied Volatility for Volatility Forecasting." *Journal of Forecasting* 35: 167–178.

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